

overlap_minimization_map

Overlap Minimization Map

Goal: preserve shared lineage while preventing each preprint from repeating the same mathematics, architecture claims, or implementation assertions.

Shared Text Budget

Each paper may reuse:

- One paragraph defining Spatium Computationis v3.
- One paragraph placing the paper within the four-part series.
- One short source-lineage note.
- One compact limitations paragraph referencing evidence gaps.

Each paper should avoid:

- Copying the full original abstract.
- Repeating all four canonical contributions.
- Reusing the same conclusion claims.
- Restating all equations from other papers.

Paper-Specific Ownership

Material	Owner paper	Allowed cross-reference
Velocity Verlet equations	Paper 1	Paper 4 may cite as substrate, no derivation.
N-body gravitational and Coulomb force laws	Paper 1	Paper 3 may mention particle dynamics only.
Relativistic speed bound	Paper 1	Paper 4 may cite as causality constraint.
Conservation-law diagnostics	Paper 1	Papers 2-4 may cite validation norm.
Metric tensor and Christoffel symbols	Paper 2	Paper 3 may cite metric substrate.
Geodesic equation and RK4 integration	Paper 2	Paper 4 may cite geometric routing conceptually.
Curvature tensors and invariants	Paper 2	Paper 3 may mention topology/curvature at high level.
Horizon detection and parallel transport	Paper 2	Paper 3 may cite horizon entropy only.

Material	Owner paper	Allowed cross-reference
Topology catalog	Paper 3	Paper 2 may list benchmark metrics without sandbox framing.
Friedmann evolution	Paper 3	Paper 2 may mention FLRW as a metric example only.
Entropy and inter-verse channels	Paper 3	Paper 4 may cite sandbox boundaries only.
Three-tier signal classification	Paper 4	No other paper should include details.
Agent coordinates and zones	Paper 4	Papers 1-3 should not include them.
Security/battleground semantics	Paper 4	No other paper should include details.

Abstract Differentiation

Paper 1 abstract vocabulary:

- numerical mechanics
- symplectic integration
- conservation diagnostics
- force-mediated simulation

Paper 2 abstract vocabulary:

- metric functions
- numerical differential geometry
- geodesic computation
- curvature analysis

Paper 3 abstract vocabulary:

- mini-verse
- configurable constants
- topology
- Friedmann evolution
- entropy and inter-verse boundaries

Paper 4 abstract vocabulary:

- autonomous agents
- spatial embedding
- routing
- quarantine
- operational zones

Conclusion Differentiation

- Paper 1 concludes with a validation-first mechanics substrate.
- Paper 2 concludes with geometry as a runtime-modifiable computational layer.

- Paper 3 concludes with sandbox design and experiment families.
- Paper 4 concludes with spatial architecture for agent coordination and release-gated security semantics.

Shared Lineage Citation Pattern

Recommended internal citation note:

This paper derives from the canonical Spatium Computationis v3 packet, version 3.0, prepared in June 2026. It isolates one contribution surface from the broader packet to reduce overlap and clarify evidence requirements.

Anti-Overlap Reviewer Checklist

- Does the paper make one primary contribution?
- Are equations copied only when they are owned by the paper?
- Are implementation claims stated only where evidence is expected?
- Does the abstract avoid listing all four series contributions?
- Does the conclusion avoid claiming the whole system is validated?
- Are private or security-sensitive details omitted unless this is Paper 4 and release-approved?